

transition to this level leaving holes behind in the valence band.

3 (a) (i) From the graph, $T = 4 \times 10 \times 10^{-3} = 0.040 \text{ s}$.

$$\text{Therefore, } f = \frac{1}{0.040} = 25 \text{ Hz.}$$

(ii) From the graph, $V_o = 3.4 \times 5.0 = 17 \text{ V}$.

(b) (i) Since $R = \frac{V_{r.m.s.}}{I_{r.m.s.}}$, we have

$$500 = \frac{\frac{17}{\sqrt{2}}}{I_{r.m.s.}}$$

$$I_{r.m.s.} \approx 0.024 \text{ A}$$

(ii) $\langle P \rangle = \frac{V_{r.m.s.}^2}{R}$

$$\langle P \rangle = \frac{\left(\frac{17}{\sqrt{2}}\right)^2}{500}$$

$$\langle P \rangle \approx 0.289 \text{ W}$$